

# Hydro Model Architectures

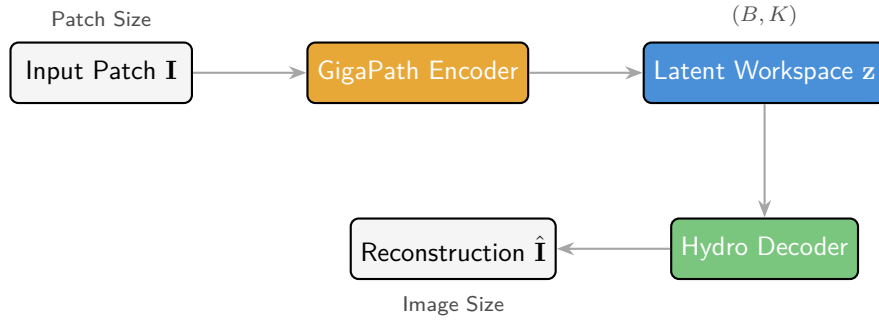
autopath

## 1 Overview

The **Hydro** datablock provides a flexible framework for latent-to-image decoders. It is designed to invert feature extractors (e.g., GigaPath) by mapping their embeddings back to the original image space.

### 1.1 Large-scale System Architecture

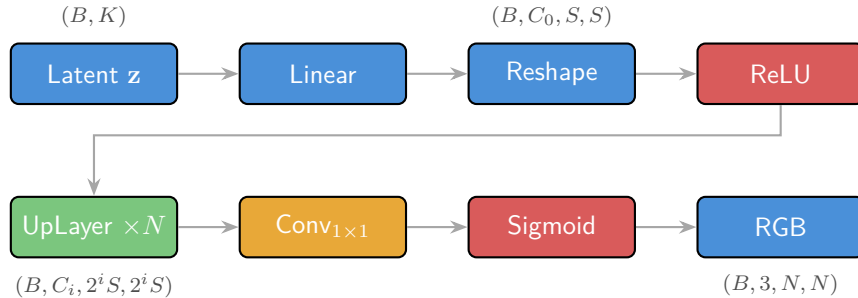
In the context of the overall pipeline, **Hydro** serves as the generative component that transforms compressed semantic embeddings into spatial reconstructions. The global latent vector  $\mathbf{z}$  is the primary interface between the large-scale encoder and the decoder.



The latent vector  $\mathbf{z}$  is produced by a fixed (or fine-tuned) backbone that summarizes the semantic content of a histology slide or patch. This vector provides the global context used by **Hydro** features like Feature Modulation to inform the progressive upsampling process.

## 2 Model\_0: Deconvolutional Decoder

**Model\_0** is the baseline architecture, using progressive upsampling to reconstruct  $N \times N$  RGB images from  $K$ -dimensional latent vectors.



## 2.1 Architectural Improvements to Combat Blur

To address the blurriness observed in `Model_0` outputs, several optional components can be enabled. These features prioritize the preservation and enhancement of high-frequency information and are situated within the progressive upsampling loop.

### 2.1.1 Overview of Improved Upsampling Path

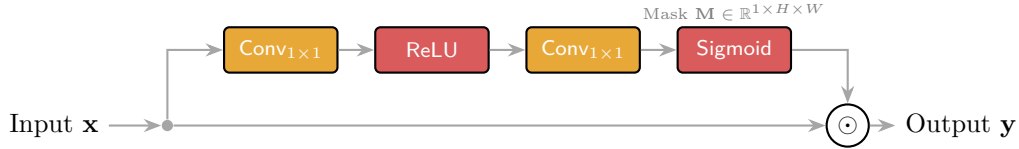
The decoding process begins with a global latent vector  $\mathbf{z} \in \mathbb{R}^K$ , which represents a compressed semantic summary of the input slide (extracted via a large-scale encoder like GigaPath). When enabled, the optional components are applied sequentially after each base `UpLayer`. The data flow for a single upsampling stage  $i$  is:

$$\mathbf{x}_{i-1} \xrightarrow{\text{Residual/UpLayer}} \hat{\mathbf{x}}_i \xrightarrow{\text{Modulation}(\mathbf{z})} \tilde{\mathbf{x}}_i \xrightarrow{\text{Attention}} \mathbf{x}_i$$

where  $\mathbf{x}_i \in \mathbb{R}^{C_i \times H_i \times W_i}$ .

### 2.1.2 Spatial Attention Gates (SAG)

The `SpatialAttentionGate` identifies and emphasizes salient spatial features (e.g., edges and textures). It is situated at the end of each upsampling stage to refine the newly generated spatial details.



The gate receives the upsampled feature map  $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$ . While the gate itself employs  $1 \times 1$  (pointwise) convolutions, it effectively learns to identify spatial transitions by interpreting the local neighborhood information that has already been “collapsed” into the channel dimension by the preceding  $3 \times 3$  convolutions in the `UpLayer`.

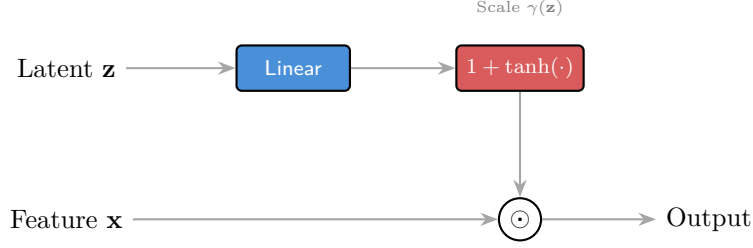
The mask generator sub-network learns a “pixelwise” spatial mask  $\mathbf{M}$  where each value  $\in [0, 1]$ . This single-channel mask is shared across all  $C$  channels. The output is computed via the Hadamard product (element-wise multiplication  $\odot$ ) with broadcasting:

$$\mathbf{y}_{c,h,w} = \mathbf{x}_{c,h,w} \times \mathbf{M}_{1,h,w}$$

By focusing representational capacity on these spatially-encoded transitions, SAG effectively reduces the smoothing artifacts.

### 2.1.3 Feature Modulation

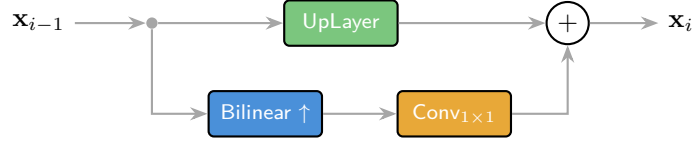
Feature modulation allows the global latent vector  $\mathbf{z}$  to dynamically scale the intensity of intermediate feature maps. This provides a mechanism for the latent representation to globally influence spatial details at every upsampling stage.



This global conditioning ensures that the reconstructed image maintains sharpening consistency across different scales, as the latent vector can “boost” specific frequency components needed for a sharp reconstruction.

#### 2.1.4 Residual Upsampling

Residual upsampling incorporates long-range skip connections around the upsampling layers. This allows the network to learn only the residual difference needed to reach the next resolution, facilitating better gradient flow and preserving high-frequency details from lower spatial resolutions.



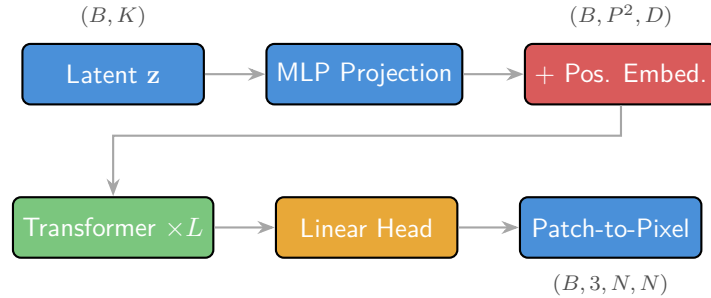
By bypassing the non-linear transformations for a portion of the signal, the model avoids the smoothing effect of repeated convolutions, resulting in crisper results.

### 3 Model\_1: Vision Transformer (ViT) Decoder

Model\_1 adopts a transformer-based architecture to model long-range dependencies and produce sharper reconstructions by operating on a tokenized representation of the image.

#### 3.1 Architecture Overview

The architecture follows a “latent token to patch” approach:



### 3.1.1 Tokenization and Projection

The latent vector  $\mathbf{z}$  is projected into a sequence of  $M = (N/p)^2$  tokens, where  $p$  is the patch size (e.g., 16).

$$\mathbf{z} \xrightarrow{\text{MLP}} \mathbf{T} \in \mathbb{R}^{M \times D}$$

Learnable 2D positional embeddings are added to provide spatial context to the transformer.

### 3.1.2 Transformer Backbone

A series of  $L$  standard Transformer blocks (Multi-Head Self-Attention + MLP) process the token sequence. This allows information to propagate globally across the image grid.

### 3.1.3 Reconstruction Head

Each token  $\mathbf{t}_i \in \mathbb{R}^D$  is projected to a patch  $\mathbf{P}_i \in \mathbb{R}^{3 \times p \times p}$  using a linear layer. The patches are then rearranged to form the final image  $\mathbf{I} \in \mathbb{R}^{3 \times N \times N}$ .

## 4 System Configuration

The Hydro datablock uses a shared configuration dataclass:

| Parameter                   | Default | Description                                            |
|-----------------------------|---------|--------------------------------------------------------|
| latent_dim                  | 1536    | Input latent vector dimension $K$                      |
| image_size                  | 256     | Output resolution $N \times N$                         |
| hidden_channels             | 256     | Base channel count (Model_0) / Dimension $D$ (Model_1) |
| patch_size                  | 16      | Patch size for Model_1                                 |
| num_layers                  | 12      | Number of layers (Transformer blocks) for Model_1      |
| version                     | 0       | Model architecture version (0 or 1)                    |
| use_residual_upsampling     | False   | Enable residual skip connections in Model_0            |
| use_spatial_attention_gates | False   | Enable spatial attention in Model_0                    |
| use_feature_modulation      | False   | Enable latent-based feature modulation in Model_0      |
| loss_type                   | mse     | MSE, L1, L0, or LPIPS reconstruction loss              |

## 5 Loss Functions

Reconstruction quality is improved by incorporating perceptual metrics:

$$\mathcal{L} = \begin{cases} \frac{1}{3N^2} \|\hat{\mathbf{I}} - \mathbf{I}^*\|_2^2 & \text{if loss\_type = "mse"} \\ \frac{1}{3N^2} \|\hat{\mathbf{I}} - \mathbf{I}^*\|_1 & \text{if loss\_type = "l1"} \\ \text{LPIPS}(\hat{\mathbf{I}}, \mathbf{I}^*) & \text{if loss\_type = "lpips"} \end{cases}$$

The **LPIPS** (Learned Perceptual Image Patch Similarity) loss compares deep features from a pre-trained network (e.g., VGG) and is highly effective at reducing blur by penalizing differences in perceptual content rather than raw pixel values.